2D Array:- Search in 2D Matrix :-

$$\begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

(row = 0, col = m-1)

target = 6

n → row number → matrix.length

m → col number → matrix[0].length

mat[][];

9

2

$$\begin{bmatrix} \\ \end{bmatrix}_{2\times 3}$$

(Leetcode 240)

```
row=0, col=m-1;
while(row<n && col>=0){
    if(mat[row][col]==target)
        return true;
    else if(mat[row][col]>target)
        col--;
    else {
        row++;
    }
}
```

mat[row][col] < target

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$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 4 & 5 & 6 & 7 \\ 8 & 9 & 10 & 11 \end{bmatrix}$$

$$[\,0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \quad 9\,]$$

$\frac{5}{4} = 1$

row = mid / m

col = mid % m

11 → 3, 4

5 % 4 = 1

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 10 & 11 & 16 & 20 \\ 23 & 30 & 34 & 60 \end{bmatrix}$$

11 → mid

(target = 3)

(suppose mid is here)

(,

1	3	5	7	10	11	16	20	23	30	34	60
0	1	2	3	4	5	6	7	8	9	10	11

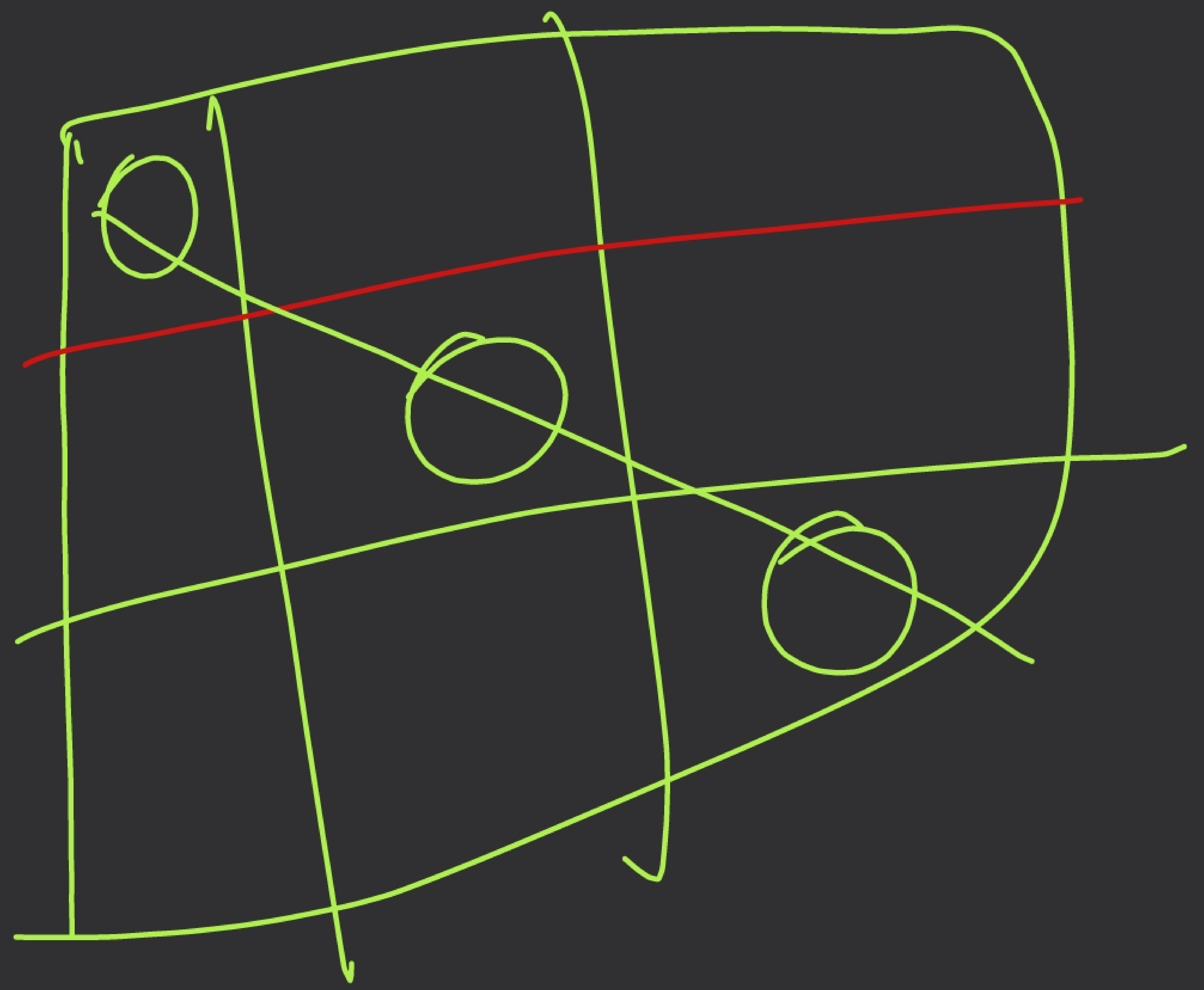
(mid)

arr[mid]

5 → mid → 1, 1

60 → 11/4 = 2, 11 % 4 = 3

```
int start = 0, end = n*m-1;
while (start <= end) {
    int mid = (start + end)/2;
    int rowIndex = mid/m;
    int colIndex = mid % m;
    if (mat[rowIndex][colIndex] == target)  return true;
    if (    "       > target)    end = mid-1;
    else start = mid+1;
    }
}
```



$$\begin{array}{c} \begin{array}{ccc} 00 & & \end{array} \\ \begin{bmatrix} 1 & 2 & 3 \\ 4 & \overset{11}{5} & 6 \\ 7 & 8 & \overset{22}{9} \end{bmatrix} \end{array}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

```
for(i=0; i<arr.length; i++){
    int j = i;   X
    sum += mat[i][j];
        or
    sum += mat[i][i];
}
```

$$
\begin{array}{ccccc}
00 & & & & 04 \\
\end{array}
$$

$$
\begin{bmatrix}
1 & 2 & 3 & 4 & 5 \\
6 & 7 & 8 & 9 & 10 \\
11 & 12 & 13 & 14 & 15 \\
16 & 17 & 18 & 19 & 20 \\
21 & 22 & 23 & 24 & 25
\end{bmatrix}
$$

11 13 22 31 33 40 44

[1, 2,]

$5 \times 5 \rightarrow 5$

su
int diagias = n - 1;

66

33 → $\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

22

$n = 5$

mat[i][i]

Secondary → $[i][i] \longleftrightarrow [i][n-i-1]$

$[1][1] \longleftrightarrow [1][3]$

$[i][i] \longleftrightarrow [i][n-i-1]$

mat$[i][n-i-1]$

i 4

$[0][0] \longleftrightarrow [0][4] \rightarrow (5-0-1)$

$[1][1] \longleftrightarrow [1][3] \rightarrow (5-1-1)$

$[2][2] \longleftrightarrow [2][2] \rightarrow (5-2-1)$

$[3][3] \longleftrightarrow [3][1] \rightarrow (5-3-1)$

$[4][4] \longleftrightarrow [4][0] \rightarrow (5-4-1)$

$5 \% 2 == 1$

$(n \% 2 == 1)$

↓

sum = sum − mat$\left[\frac{n}{2}\right]\left[\frac{n}{2}\right]$;

// code for all unique diagonal element

```
for(i=0; i<n; i++){
    if(mat[i][i] == mat[i][n-i-1])
        -> sum += mat[i][i];
    else{
        sum += mat[i][i];
        sum += mat[i][n-i-1];
    }
}
return sum;
}
```

Leetcode 1572 :-

```
int n = arr.length;
for (i = 0 → n) {
    sum = sum + arr[i][i];        → //primary diagonal
    sum = sum + arr[i][n-i-1];    → // secondary
}
// if it is odd matrix
if (n % 2 == 1) {
    sum = sum - arr[n/2][n/2];
}
return sum;
}
```

H.W.

→ Move zeroes to end

→ Trapping Rain Water